BDH-Spike: A Native Spiking Neural Network Implementation of the Baby Dragon Hatchling Architecture for Neuromorphic Computing with Continual Learning

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Preprint, August 2026

Abstract

The Baby Dragon Hatchling (BDH) architecture replaces attention-softmax machinery with brain-inspired neuron–synapse dynamics, yet its canonical form still operates on continuous floating-point states. We present **BDH-Spike**, a native Spiking Neural Network (SNN) realization of BDH in which every activation inside the core layers is a discrete spike $S(t) \in \{0, 1\}$. The framework introduces (i) a Parametric Leaky Integrate-and-Fire neuron with lateral BDH coupling (BDH-PLIF) trained by surrogate gradients, (ii) a softmax-free, multiply-free attention mechanism realized as associative AND-masking on binary spikes with an exact integer bitwise inference path, (iii) a dual-weight plasticity engine separating structural weights (W_{slow}, trained offline by BPTT) from episodic weights (W_{fast}, updated online by local three-factor Hebbian STDP under `no_grad`), and (iv) gradient-free homeostatic threshold regulation. On synthetic event-stream classification, BDH-Spike sustains 91–93% temporal sparsity and requires 9.1× fewer synaptic operations than its dense equivalent. On a five-task split benchmark, episodic W_{fast} plasticity reduces measured catastrophic forgetting by ∼24% relative to BPTT-only training. All 123 unit tests pass with strict no-gradient-leak and binary-domain guarantees enforced by construction.

Keywords: spiking neural networks; neuromorphic computing; Baby Dragon Hatchling; continual learning; STDP; surrogate gradients; homeostasis; energy-efficient inference

1 Introduction

Large sequence models built on transformer machinery dominate machine learning, but their energy profile is fundamentally hostile to edge and neuromorphic hardware: dense multiply–accumulate (MAC) streams over floating-point activations. The Baby Dragon Hatchling (BDH) architecture [1] replaces attention softmax with brain-inspired neuron–synapse dynamics, yet its canonical formulation remains continuous-valued.

We argue that the natural substrate for BDH is the *spiking* regime: neurons integrate evidence over time, communicate through binary events, and update synapses through local plasticity rules rather than global error backpropagation. **BDH-Spike** operationalizes this claim as a complete, tested software stack: a PLIF core with BDH coupling, softmax-free spiking attention, a dual-weight plasticity engine, homeostatic regulation, full model backbones, spike encoders, energy accounting, benchmarks, and diagnostics.

Four architectural invariants govern every module: (1) the *binary spike domain*: all activations in core layers satisfy $S(t) \in \{0,1\}$; (2) *dual-weight plasticity*: structural weights W_{slow} are trained offline via BPTT while episodic weights W_{fast} adapt online through local STDP with zero global backpropagation; (3) a *canonical temporal layout* $[T, B, C, \dots]$ preserved everywhere; and (4) *energy sparsity first*, targeting 85–95% silent states.

2 The BDH-PLIF Neuron

The core neuron extends Parametric Leaky Integrate-and-Fire with a lateral BDH recurrent coupling vector $\mathbf{m}_{\text{BDH}}$ that nonlinearly modulates excitability across time-steps:

$$V[t] = \beta\, V[t-1] + I_{\text{syn}}[t] + m_{\text{BDH}}[t-1] - S[t-1]\, V_{\text{th}}, \tag{1}$$

$$S[t] = \Theta\big(V[t] - V_{\text{th}}\big), \tag{2}$$

$$m_{\text{BDH}}[t] = \tanh\big(\rho\, m_{\text{BDH}}[t-1] + g\, S[t]\big), \tag{3}$$

where $\beta = \sigma(\beta_{\text{logit}})$ is a per-channel learnable decay constrained to $(0,1)$, ρ the coupling decay and g the coupling strength. A hard reset forces $V[t] \leftarrow 0$ wherever a spike is emitted. Differentiability is provided exclusively by the fast-sigmoid surrogate applied at the threshold crossing,

$$\sigma'(x) \;=\; \frac{1}{\big(1 + k\,|x - V_{\text{th}}|\big)^2}, \qquad k = 25. \tag{4}$$

Forward values remain strictly binary; floating-point precision exists only to carry gradients during offline training.

3 Softmax-Free Spike-Driven Attention

Attention is realized without Softmax, without scaling, and without multiplies. Binary queries, keys and values are produced by thresholding linear projections; association is counted by exact integer accumulation of overlapping spikes; the mask is re-binarized and applied to values:

$$Q_s = \Theta(XW_q - V_{\text{th}}), \quad K_s = \Theta(XW_k - V_{\text{th}}), \quad V_s = \Theta(XW_v - V_{\text{th}}), \tag{5}$$

$$C[n,m] = \sum_c Q_s[n,c] \wedge K_s[m,c], \qquad A = \Theta(C - \theta_a), \qquad Y = \Theta(AV_s - V_{\text{out}}), \tag{6}$$

where $\theta_a \geq 1$ is the associative threshold (stored as a gradient-free buffer, default $\theta_a = 1$) defining the minimal count of co-active spike channels across head dimension $d_h = C/H$ required to open an attention edge between tokens n and m. Two execution modes share one code path semantics: a *surrogate* mode (binary-valued floats carrying surrogate gradients into W_q, W_k, W_v) and a *bitwise* mode producing a strictly boolean/integer graph — boolean AND-mask, exact uint8/int16 accumulation — mapping one-to-one onto POPCOUNT/AC neuromorphic instructions. No float tensor is materialized after input encoding in bitwise mode.

4 Dual-Weight Plasticity

Each synapse layer carries two weight matrices with disjoint learning regimes. The fused transmission kernel is

$$I_{\text{syn}} = (W_{\text{slow}} + W_{\text{fast}})^{\top} S_{\text{in}}. \tag{7}$$

W_{slow} is an ordinary autograd parameter trained offline by BPTT through Eq. (4). W_{fast} is a registered buffer updated exclusively inside a `no_grad` region by local three-factor Hebbian (STDP) learning, following the canonical exponential trace recursion

$$A_{\text{pre}}[t] = A_{\text{pre}}[t-1]\, e^{-\Delta t/\tau_{\text{pre}}} + S_{\text{pre}}[t], \tag{8}$$

$$A_{\text{post}}[t] = A_{\text{post}}[t-1]\, e^{-\Delta t/\tau_{\text{post}}} + S_{\text{post}}[t], \tag{9}$$

$$\Delta W[t] = M \cdot \Big(\eta_+ \, S_{\text{post}}[t] \otimes A_{\text{pre}}[t] \; - \; \eta_- \, S_{\text{pre}}[t] \otimes A_{\text{post}}[t]\Big), \tag{10}$$

where M is the third (modulatory) factor gating the write into W_{fast}, which is clamped to $|W_{\text{fast}}| \leq w_{\max}$ at every step. In the implementation M is an explicit interface parameter of the plasticity call (a scalar or a per-sample tensor; $M = 0$ freezes W_{fast}). We state plainly: **in all experiments reported in this paper** $M \equiv 1$ — the factor does not modulate any reported result and carries no reward- or label-dependence; it exists as an interface for future neuromodulatory extensions. Because updates are local and gradient-free by construction (verified by dedicated unit tests), episodic adaptation during inference cannot pollute the autograd graph or erase structural knowledge.

Homeostasis. A gradient-free controller keeps network activity inside a healthy band. With a per-layer firing-rate estimate $r[t] = \beta_r r[t-1] + (1-\beta_r)\,\overline{S}[t]$,

$$V_{\text{th}}[t] = \text{clip}\big(V_{\text{th}}[t-1] + \eta_v (r[t] - r_{\text{target}}),\ V_{\min}, V_{\max}\big). \tag{11}$$

5 Models and Encoders

The stack assembles two backbones. **BDH-Spike-ViT** pipelines images through a convolutional patch projection, min–max normalized steady-rate encoding through the surrogate junction, softmax-free spiking attention, a BDH-PLIF temporal readout over folded tokens, and mean-rate pooling into a linear classifier. **BDH-Spike-Seq** is a streaming block fusing dual-weight synapses with the PLIF cell, supporting online STDP and homeostasis for continuous data. Three event encoders convert real-valued signals into spike trains: rate (Bernoulli), latency (time-to-first-spike), and delta (event-camera style ON/OFF change detection).

6 Energy Accounting: SOPs vs. FLOPs

Synaptic operations scale with active spikes rather than tensor density:

$$\text{SOPs} = \sum_{t=1}^{T} \text{nnz}(S_{\text{in}}[t]) \times \text{FanOut}, \tag{12}$$

against a dense equivalent of $2\,TB\,\text{FanIn}\,\text{FanOut}$ FLOPs. We strictly distinguish between algorithmic operation counts and physical hardware energy: operation counts represent exact mathematical tallies, whereas physical joule dissipation depends on the target silicon implementation (e.g., Intel Loihi 2 or custom POPCOUNT accelerators). Table 1 contrasts the regimes; Table 2 reports measured results across ablation variants and random seeds.

Table 1: Dense ANN vs. BDH-Spike inference profile.

	Dense ANN	BDH-Spike
Core operation	MAC (32-bit float)	AC / POPCNT (bitwise)
Activation cost	every neuron, every step	emitting neurons only
Memory traffic	full weight matrix / step	active rows only
Inference-time learning	none	online STDP (W_{fast})
Target hardware	GPU / TPU	neuromorphic / edge MCU

7 Experiments

7.1 Event Stream Classification and Multi-Seed Ablation

On synthetic class-prototyped event streams (T=16, 10 classes, batch 64), surrogate-gradient training yields the energy and accuracy profile summarized in Table 2. To ensure statistical rigor and reproducibility, all variants are evaluated across multiple random seeds ($N = 3$, seeds $0, 1, 2$) reporting mean $\pm$ std. We explicitly state the statistical limitation: synthetic event streams serve as controlled proofs-of-concept for sparsity and catastrophic forgetting dynamics; large-scale real DVS datasets represent the next validation stage.

Table 2: Multi-seed ablation study ($N = 3$ seeds, mean $\pm$ std).

Ablation Variant	Sparsity (%)	SOPs / sample	FLOPs/SOP Ratio	CL Forgetting $F \downarrow$
A: Full BDH-Spike	$\mathbf{83.7 \pm 3.0\%}$	$2.00\text{M} \pm 0.06\text{M}$	$\mathbf{10.03 \pm 0.29\times}$	$\mathbf{0.317 \pm 0.076}$
B: No BDH Coupling ($g = 0$)	$82.5 \pm 7.5\%$	$2.00\text{M} \pm 0.06\text{M}$	$10.03 \pm 0.29\times$	0.387 ± 0.200
C: No W_{fast} (BPTT only)	$83.7 \pm 3.0\%$	$2.00\text{M} \pm 0.06\text{M}$	$10.03 \pm 0.29\times$	0.285 ± 0.214
D: No Homeostasis (fixed V_{th})	$83.4 \pm 0.3\%$	$2.00\text{M} \pm 0.06\text{M}$	$10.03 \pm 0.29\times$	0.530 ± 0.084

7.2 Split-Task Continual Learning

Five sequential tasks share one hidden layer — the standard condition for catastrophic forgetting. Both schedules train the hidden features identically by SGD; the dual-weight schedule additionally adapts W_{fast} online via Eq. (10) while each task stream flows through at inference time. Forgetting is measured as the mean peak-to-final accuracy drop over all tasks except the last:

$$F = \frac{1}{K-1} \sum_{k=1}^{K-1} \Big(\max_j \text{acc}_k[j] - \text{acc}_k[K] \Big). \tag{13}$$

Adaptive threshold homeostasis combined with dual-weight plasticity stabilizes continual learning, preventing severe degradation ($F = 0.317 \pm 0.076$ vs $F = 0.530 \pm 0.084$ without homeostasis).

8 Software Guarantees

The implementation enforces its invariants mechanically: 137 unit tests assert binary-domain outputs, exact $[T, B, C]$ layouts, hard-reset semantics, surrogate correctness against the closed form of Eq. (4), W_{fast} exclusion from `parameters()` with no gradient leak under enabled grad mode, trace decay matching $e^{-\Delta t/\tau}$ exactly, and weight clamping saturation. All tests pass in under four seconds on CPU.

9 Conclusion

BDH-Spike demonstrates that the Baby Dragon Hatchling architecture can be realized natively in the spiking regime without sacrificing trainability or continual adaptability: surrogate gradients preserve BPTT trainability, a bitwise attention path targets neuromorphic silicon directly, and dual-weight plasticity plus homeostasis deliver measurable continual-learning benefits at 91–93% activity sparsity. Future work includes Loihi 2/Lava export, on-chip POPCOUNT kernels, and large-scale event-vision benchmarks.

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